

An obstruction theoretic approach to integrated information

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Abstract

We define the integrated cohomology groups of physical systems, and show that integrated information can be detected by the non-vanishing of H^1 obstruction classes that measures the failure of the cause-effect properties of the system to decompose into smaller parts. This gives a completely functorial notion of integrated information, as suggested by Tull–Kleiner, and allows us to further define higher order integration in a very natural way.

1 Introduction

Integrated Information Theory (IIT) is a mathematical description of the existing consciousness of a system, developed mainly by Tononi and his collaborators, [CITE](#). The theory is centered around the computation of Φ , which is a measure of the *integrated information* in the system. Informally, a system S in a state s , has integrated information, captured by $\Phi(S, s) > 0$, if the information generated by the entire system’s cause-effect structure in state s is greater than the information generated by the combined cause-effect structure of partitions of the system. The formal mathematical framework for computing Φ is rigorous and technical, and uses detailed features of probability theory and measure theory, see [CITE](#)

In the present paper we wish to study integrated information through the lens of obstruction theory, which is a mathematical framework studying certain obstruction classes that measures whether a property can be extended through defined levels of sub-structure. They live in mathematical objects called *cohomology groups*, which are abstract formulations of the amount of holes a system has in different dimensions. Obstruction theory was originally developed by Stiefel and Whitney to prove the non-existence of certain vector-fields on man-

ifolds [CITE](#), but has later been applied in a plethora of mathematical contexts [CITE](#).

The conceptual similarity between these two ideas is also extendable to a formal description, which is the central aim of this short paper. Building on work by Kleiner and Tull on a categorical description of integrated information theory, see [CITE](#), we assign a simplicial complex $N^\bullet$ to an IIT system S , and use this to build cohomology groups valued in a sheaf $\mathcal{F}$ that incorporates the cause-effect dynamics of S . Intuitively, we wish to build a mathematical structure whose n -dimensional holes represent n -th order integration, or causal connectedness. We then define a natural obstruction class Δ^ψ for any attempted partition ψ of S into subsystems, which we show implies properties for IIT. In particular, if there is a non-trivial obstruction class, then the system has integrated information. These classes will detect whether the global cause-effect dynamics of S can be “glued together” from the local dynamics of its decompositions. This gluing is described by the presheaf, and the failure to glue is measured by the cohomology.

Theorem 1.1. *Let s be a state of a system S . If the obstruction classes $\Delta^\Psi \in H^1(N^\bullet; \mathcal{F})$ vanish, then $\Phi(S, s) > 0$.*

The upshot of this work is that one can study integrated information from a new lens, and apply fundamental ideas from algebraic topology and homological algebra to study phenomena in consciousness. In particular, our theory gives a much more general approach to integrated information, that side-steps a lot of technicalities. We expect that our methods can be combined with topological data analysis, see [CITE](#) for an introduction, to locate minimum partitions (MIP) and connect integrated information to new developments in the relationship between topology and information theory, see for example [CITE](#) and [CITE](#).

In the last section we consider two natural extensions of our framework: the size of Φ and higher order in-

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tegration. In particular, we construct a cohomological graded integrated information value Φ_{coh}^k , which we conjecture is related to classical integrated information Φ . Our above theorem implies that $\Phi = 0 \implies \Phi_{coh}^1 = 0$.

We make no claim surrounding the validity of IIT to explain consciousness and the properties of experience, and make no contributions to the ongoing debates [CITE](#). We only wish to connect some of IIT's mathematical properties to other mathematical concepts known to measure similar features. We view this work not as a new standard for IIT, but as a starting point to explore some of the rich mathematical structures related to it.

2 IIT systems

In [CITE](#), Tull and Kleiner develop a category theoretic foundation for integrated information theory, based on symmetric monoidal categories with some extra structure that allows for the modeling of cause-effect structures. We will use this as a backdrop for our cohomology theory, hence we quickly review their setup.

2.1 Process theories and decompositions

Definition 2.1. A *process theory* is a symmetric monoidal dagger category $(\mathcal{C}, \otimes, \mathbb{1}, \dagger)$, usually denoted simply by $\mathcal{C}$, together with two maps $q_S: S \rightarrow \mathbb{1}$ and $w_S: \mathbb{1} \rightarrow S$ for any $S \in \mathcal{C}$ satisfying monoidality and standard naturality conditions. Objects in $\mathcal{C}$ are interpreted as system types, and morphisms as physical processes. The dagger $\dagger: \mathcal{C}^{op} \rightarrow \mathcal{C}$ is interpreted as a time-reversal operation.

Morphisms of the form $\mathbb{1} \rightarrow S$ will be referred to as *states*, and we will write $\text{St}(S)$ for the collection of states of S . Similarly, morphisms $S \rightarrow \mathbb{1}$ will be referred to as *effects* and morphisms $\mathbb{1} \rightarrow \mathbb{1}$ *scalars*. The designated maps q and w are called the *discarding effect* and the *noise state* respectively, and are interpreted as such.

Definition 2.2. A morphism $f: S \rightarrow S'$ is *causal* if it preserves the discarding effect,

$$q_{S'} \circ f = q_S,$$

and similarly *cocausal* if it preserves the noise state.

Remark 2.3. Tull-Kleiner does not assume $\mathcal{C}$ to be a dagger category for their most general setup, but for concretely tying the categorical version to the IIT algorithm of [CITE](#), as we do in [Section 3.3](#), this is required.

Example 2.4. The standard example of a process theory is the category of classical probabilistic processes. This is the category Class of finite sets, where a morphism $S \rightarrow S'$ assigns to an element $s \in S$ an 'unnormalized probability distribution' of elements in S' , which are maps $S \times S' \rightarrow \mathbb{R}^+$. The time reversal operation is given by $f^\dagger(s, s') = f(s', s)$. Composition of two maps $f: S \times S' \rightarrow \mathbb{R}^+$ and $g: S' \times S'' \rightarrow \mathbb{R}^+$, is defined by

$$f \circ g: S \times S'' \rightarrow \mathbb{R}^+ \\ (s, s'') \mapsto \sum_{s' \in S'} f(s, s')g(s', s'').$$

The monoidal structure is the cartesian product, meaning the unit $\mathbb{1}$ is the singleton set. Product of morphisms is given by

$$f \otimes g(s, s'')(s', s''') = f(s, s')g(s'', s''').$$

The discarding process q_S is the unique process with $q_S(s) = 1$ for all $s \in S$, meaning a process is causal whenever it is stochastic. The noise state is the uniform probability distribution with $w_S(s) = \frac{1}{|S|}$.

Definition 2.5. Let $\mathcal{C}$ be a process theory, and S an object in $\mathcal{C}$. A *decomposition* of S is a pair of objects $A, B \in \mathcal{C}$ together with a causal isomorphism $\psi: S \simeq A \otimes B$. We denote decompositions by (A, B, ψ) , or sometimes simply by (A, B) .

Any object $S \in \mathcal{C}$ has a decomposition $(S, \mathbb{1})$, which we call the *trivial* decomposition. We will further say that two decompositions (A, B, ψ) and (A', B', ψ') are *equivalent* if there are causal isomorphisms $f: A \simeq A'$ and $g: B \simeq B'$, such that

$$\psi' = (f \otimes g) \circ \psi.$$

The *complement* decomposition of (A, B) is the same decomposition, just in the opposite order; it is denoted $(A, B)^\perp = (B, A)$. Note that, if two decompositions are equivalent, then so are their complements.

Definition 2.6. Let $\mathbb{D}(S)$ denote the set of equivalence classes of decompositions of S . The operation $(-)^\perp$ acts on $\mathbb{D}(S)$, and we define a *decomposition set* of S to be a subset $\mathbb{D} \subseteq \mathbb{D}(S)$ that contains the trivial decomposition and is closed under $(-)^\perp$.

Definition 2.7. Given a decomposition set $\mathbb{D}$ of S , and a decomposition $(A, B, \psi) \in \mathbb{D}$, we define the *restricted* decomposition set $\mathbb{D}|_A$ to be the set of decompositions of A that can be extended to a decomposition of S in $\mathbb{D}$. Furthermore, we define the *restriction* of a state $s: \mathbb{1} \rightarrow S$ to A to be the state defined by

$$s|_A = (\text{Id}_A \otimes q_B) \circ \psi \circ s: \mathbb{1} \rightarrow A,$$

and similarly for B .

Intuitively, the restriction is defined by identifying S with its decomposition, and then discarding the factor we don't want to consider. By functoriality and naturality, such restrictions depend only on the equivalence class in $\mathbb{D}(S)$, and not on the particular representative.

2.2 System types and directional cuts

To define generalized integrated information theories, we need a notion of systems to study. These are objects in the ambient category $\mathcal{C}$ together with the extra structure defined in the previous section.

Definition 2.8. A *system* in $\mathcal{C}$ is a triple $(S, \mathbb{D}, T)$, where $S \in \mathcal{C}$, $\mathbb{D}$ is a decomposition set of S , and $T: S \rightarrow S$ is a causal process, interpreted as a *time evolution*. It is assumed that also $T^\dagger$ is causal.

Example 2.9. The most basic example is the *trivial* system $(\mathbb{1}, \{\mathbb{1} \otimes \mathbb{1}\}, \text{Id}_{\mathbb{1}})$.

A state of a system $(S, \mathbb{D}, T)$ is a state of S . Given such a state s and a decomposition $(A, B) \in \mathbb{D}$, **CITE** constructs a *subsystem* $(A, \mathbb{D}|_A, T|_A)$, where the restricted time evolution is defined by the composition

$$A \xrightarrow{1 \otimes s|_B} A \otimes B \xrightarrow{T} A \otimes B \xrightarrow{1 \otimes q_A} A.$$

These subsystems are again independent of the class representative in $\mathbb{D}$.

Given a system $(S, \mathbb{D}, T)$ and a decomposition $(A, B, \psi) \in \mathbb{D}$, one needs to be able to form systems that disconnects certain causal connections in the total system. IIT 3.0 **CITE** does this by introducing *cut systems*, that directionally disconnects causal flow from the two parts. IIT 4.0 **CITE** does this in a slightly different manner, by replacing parts of the inputs of the time evolutions with independent noise. We have included versions of IIT 4.0's disconnections in arbitrary system states for reference and intuition, but we

remark that the actual implementation of such disconnections does not matter much, as long as they are functorial.

Definition 2.10. Given a decomposition $(A, B, \psi) \in \mathbb{D}$, the *left cut* replaces the contribution of B to the time evolution T with the noise state, meaning that only the contribution from A matters. Formally, it is the system $(S, \mathbb{D}, T^{\psi, \leftarrow})$, where the new time evolution operator is defined as

$$T^{\psi, \leftarrow}: A \otimes B \xrightarrow{1 \otimes w_B q_B} A \otimes B \xrightarrow{T} A \otimes B.$$

In spirit this forces the time evolution to retain all self-influences from A , and influences from A to B , but removes all influences from B to A . The arrow in the notation is meant to denote the direction that is removed.

We define the *right cut* similarly by replacing T with

$$T^{\psi, \rightarrow}: A \otimes B \xrightarrow{w_A q_A \otimes 1} A \otimes B \xrightarrow{T} A \otimes B,$$

and the *double cut* by

$$T^{\psi, \leftrightarrow}: A \otimes B \xrightarrow{1 \otimes w_A \otimes w_B \otimes 1} A \otimes B \otimes A \otimes B \xrightarrow{T \otimes T} A \otimes B \otimes A \otimes B \xrightarrow{1 \otimes q_B \otimes q_A}$$

Remark 2.11. The double cut corresponds to Tull-Kleiner's bidirectional cut, where one evolves the system as if A and B were completely disconnected from each other.

We will use the notation $T^{\psi, \alpha}$, with $\alpha \in \{\leftarrow, \rightarrow, \leftrightarrow\}$ to denote a general disconnection based on a decomposition $\psi: S \simeq A \otimes B$. As we did above with the restricted time evolution, we also get definitions of restricted cuts due to the functoriality of discarding and noise; these are denoted $T|_A^{\psi, \alpha}$.

3 Integrated cohomology and obstruction classes

In this section we present the more novel parts of this paper, which is a cohomology theory for IIT systems. We then build an obstruction theory for decomposing a physical system based on this cohomology theory, and show that it detects the existence of integrated information. Lastly we extend this to naturally define higher order integration, often referred to as higher *synergy*.

3.1 Prerequisites

In essence, cohomology groups are mathematical constructions that measure the amount of holes of different dimensions a space or other mathematical object has. It is generally defined using homological algebra, see [CITE](#) for an introduction.

Definition 3.1. A *cochain complex* $C^\bullet$ is a sequence of abelian groups

$$\dots \longrightarrow C^{i-1} \xrightarrow{\delta^{i-1}} C^i \xrightarrow{\delta^i} C^{i+1} \longrightarrow \dots$$

such that $\delta^{i-1} \circ \delta^i = 0$ for all i .

The elements of $C^\bullet$ are called the i -cochains. An i -cochain in the image of the map δ^{i-1} is called a coboundary, while an i -cochain in the kernel of δ^i is called a cocycle. The equation $\delta^{i-1} \circ \delta^i = 0$ implies that $\text{Im}(\delta^{i-1}) \subseteq \text{Ker}(\delta^i)$, which is the fundamental fact used to define cohomology – in words, any coboundary is also a cocycle.

Definition 3.2. Given a cochain complex $C^\bullet$, its i 'th *cohomology group* is defined as the quotient group

$$H^i(C^\bullet) := \text{Ker}(\delta^i) / \text{Im}(\delta^{i-1}).$$

Elements of $H^i(C^\bullet)$ are called *cohomology classes*.

One of the central goals of algebraic topology and algebraic geometry is to functorially assign cochain complexes to topological and geometric objects and compute the resulting homology groups. For a category of interest $\mathcal{C}$, for example topological spaces, manifolds or schemes, such assignments defines functors into the category of graded abelian groups $H^*: \mathcal{C} \longrightarrow \text{Ab}_*$ called *cohomology theories*. Examples include *simplicial cohomology* [CITE](#), *de Rham cohomology* [CITE](#) and *coherent cohomology* [CITE](#).

3.2 Integrated cohomology

Given a system $(S, \mathbb{D}, T)$ and a global state $s: \mathbb{1} \longrightarrow S$, we will denote by $\text{Sub}_s(S)$ the category of subsystems of S , defined as the poset category associated to the poset structure in [CITE](#). In other words, $\text{Sub}_s(S)$ is the category consisting of subsystems as objects, and morphisms being single comparison maps by the pre-order structure.

The overall idea of our approach can be described as follows:

1. first, define a presheaf $\mathcal{F}$ on $\text{Sub}_s(S)$ that describes the cause-effect dynamics of subsystems of S ;
2. second, define the integrated cohomology of S to be the presheaf-cohomology of the nerve of $\text{Sub}_s(S)$ valued in $\mathcal{F}$;
3. third, define natural obstruction classes in $H^1(N^\bullet(\text{Sub}_s(S)); \mathcal{F})$;
4. and lastly, prove that these obstruction classes detect the existence of integrated information in S .

There are several possible choices for what presheaf $\mathcal{F}$ to use, depending on how categorical or how system specific one wants to be. If one focuses on actual IIT 4.0 constructions, then perhaps a probability based approach using intrinsic information and intrinsic difference would be a good strategy. We have, however, opted for a more categorical and general approach. This is in part to give generalized integrated information functorial properties, as discussed by Tull–Kleiner in [CITE](#).

As we want to have access to the time evolutions T and $T^{\psi, \alpha}$ for subsystems, it is natural to consider the *endomorphisms* as a natural starting position for the presheaf. As we need to be able to understand the *difference* between T and $T^{\psi, \alpha}$, we also need access to a subtraction operation. The standard way to do this is as follows.

Definition 3.3. The *dynamic presheaf* on $\text{Sub}_s(S)$ is defined on objects as the free abelian group on the associated endomorphisms,

$$\begin{aligned} \mathcal{F}: \text{Sub}_s(S)^{\text{op}} &\longrightarrow \text{Ab} \\ A &\longmapsto \mathbb{Z}[\text{End}(A)] \end{aligned}$$

Given a subsystem $U \leq A$, we define for any endomorphism $f: A \longrightarrow A$ a restriction map

$$\rho_U^A(f) = (\text{Id}_U \otimes q_V) \circ f \circ (\text{Id}_U \otimes w_V),$$

where V is the complement of U , i.e. the object such that $A \simeq U \otimes V$, which exists by the definition of a subsystem. We then linearly extended this to $\mathbb{Z}[\text{End}(A)]$, making $\mathcal{F}$ a presheaf.

Remark 3.4. Our setup is formally analogous to Abramsky–Brandenburger’s sheaf theoretic description of contextuality, see [CITE](#) and [CITE](#), but differs in

the added dynamics of the systems. However, we feel that the similarities are enticing, and that the possible interactions are worth further investigation.

Remark 3.5. One could also naturally choose to use the actual state $s|_V$ instead of the noise state w_V in the definition of the restriction maps. Doing this would mean that the obstruction theory detects state-dependent integration, and not the intrinsic integration of the system. We believe that studying what happens upon making this change is an interesting avenue of future research.

Definition 3.6 (CITE). Let $\mathcal{C}$ be a category. The *nerve* of $\mathcal{C}$ is the simplicial complex $N^\bullet(\mathcal{C})$ defined by having a 0-simplex for each object in $\mathcal{C}$, a 1-simplex for each morphism in $\mathcal{C}$, a 2-simplex for every pair of composable morphisms, and generally an k -simplex $c_0 \rightarrow c_1 \rightarrow \dots \rightarrow c_k$ for any k -tuple of composable morphisms. For $0 < i < k$ the i 'th face maps f_i are given by composing the two maps, while for $i = 0, k$ it is given by omitting the first or last object respectively. The i 'th degeneracy map d_i is given by inserting an identity map at the i 'th object.

The nerve of a category is a way of encoding how all objects in the category are related to each other via a combinatorial space. Given a presheaf $\mathcal{F}$ on the category $\mathcal{C}$, one gets a cochain complex

$$C^i(\mathcal{C}; \mathcal{F}) = \prod_{c_0 \rightarrow \dots \rightarrow c_i} \mathcal{F}(c_0),$$

where the coboundary map $\delta^k: C^k(\mathcal{C}; \mathcal{F}) \rightarrow C^{k+1}(\mathcal{C}; \mathcal{F})$ is defined on an k -cochain θ by the alternating sum

$$\delta^k(\theta) = \sum_{i=0}^{k+1} (-1)^i d_i(\theta).$$

More explicitly, on a $k + 1$ -simplex $\sigma = c_0 \rightarrow c_1 \rightarrow \dots \rightarrow c_{k+1}$ we have

$$d_0(\theta)(c_0 \rightarrow c_1 \rightarrow \dots \rightarrow c_{k+1}) = \rho_{c_0 \rightarrow c_1}(\theta)(c_1 \rightarrow c_2 \rightarrow \dots$$

and for $i > 0$,

$$d_i(\theta)(c_0 \rightarrow \dots \widehat{c_i} \rightarrow \dots \rightarrow c_{k+1})$$

where $\widehat{c_i}$ denotes the i 'th object being omitted.

Remark 3.7. The above definition of the coboundary operator is completely analogous to the boundary operator for Čech cohomology, see CITE.

Important for us will be the coboundary of a 0-cochain θ , which is given on a 1-simplex $\sigma: U \rightarrow V$ as

$$\delta^0 \theta(\sigma) = \rho_\sigma(\theta(V)) - \theta(U) = \mathcal{F}(\sigma)(\theta(V)) - \theta(U).$$

We can now define the cohomology theory that we will use to define the obstruction classes.

Definition 3.8. Given a system $(S, \mathbb{D}, T)$ and a state $s: \mathbb{1} \rightarrow S$, we define the *integrated cohomology* of $(S, \mathbb{D}, T)$ to be the cohomology of the nerve of $\text{Sub}_s(S)$ evaluated in the dynamic presheaf $\mathcal{F}$. In other words, the i 'th integrated cohomology group is defined by

$$H_{\text{int}}^i(S) = H^i(N^\bullet(\text{Sub}_s(S)); \mathcal{F}) = \text{Ker}(\delta^i) / \text{Im}(\delta^{i-1}).$$

Remark 3.9. It is important to note that this is not simply the topological nerve cohomology of $\text{Sub}_s(S)$, as this will almost always be trivial, see for example CITE. The important part here is the use of the presheaf $\mathcal{F}$, which makes this a highly non-trivial cohomology theory.

The intuition is as follows: The nerve of $\text{Sub}_s(S)$ gives us a geometric construction of *where* integrated information can be tested. It consists formally of chains of substates of S of all lengths, with topological structure that understands how these chains are related and connected. The presheaf gives us *what* is measured on every level – in our case the system's causal dynamics. The cohomology gives us information about which local dynamics that can be glued together to form a global dynamic. If gluing fails, then there is some level of the system where its global dynamics cannot be described in terms of local dynamics, which is the essence of the system having integrated information. In particular, a 1-cocycle measures the failure of a pairwise compatibility, which we use to define obstruction classes.

Definition 3.10. Given a decomposition ψ and a direction $\alpha \in \{\mathcal{F}(c_0 \leftarrow c_1), \mathcal{F}(c_1 \leftarrow c_0)\}$ the associated obstruction class is defined as the cohomology class of the 1-cochain $\Delta^{\psi, \alpha}$ defined on 1-simplicies $\sigma: U \rightarrow V$ as

$$\Delta^{\psi, \alpha}(\sigma) = \mathcal{F}(\sigma)(T|_U) - T|_V^{\psi, \alpha}.$$

The central claim of this paper is that these classes detect whether a system S has integrated information or not. Hence, checking that these classes vanish proves

that integrated information is non-zero, which might be much easier than computing Φ , which can be a very hard task – see [CITE](#).

Remark 3.11. The reader paying minute attention has perhaps noticed that mechanisms and purviews have yet to be featured in this framework. Our approach to measuring IIT is structurally somewhat different, as instead of starting with mechanisms and purviews and then aggregate local information upwards, we have the global structure as a reference and let mechanisms and purviews be derived observables. Hence they do not arise as primitive data, but as local test probes for evaluating the obstruction classes. We will, however, need to define and use these to relate our obstruction classes to the integrated information of S , as we will do in the following section.

3.3 Obstruction classes and generalized IIT

The goal of this section is to prove the following theorem.

Theorem 3.12. *Let $(S, \mathbb{D}, T)$ be a system and $s: \mathbb{1} \rightarrow S$ a state of S . If there is no integrated information in S , then all of the decomposition obstruction classes vanish. In other words, $\Phi(S, s) = 0 \implies [\Delta^{\psi, \alpha}] = 0$ for all A, ψ and α . In particular, the existence of non-vanishing obstruction classes implies $\Phi(S, s) > 0$, and hence that the classes $[\Delta^{\psi, \alpha}]$ form obstructions to the decomposition of causal structures.*

In order to do so we need to be able to compute and understand Φ . We will use the method of *generalized IIT's* as developed by Tull-Kleiner in [CITE](#), also formalized in the categorical setting in [CITE](#). This ensures that our theory works not only for classical IIT, but also quantum IIT and other variants that one would like to study, [CITE](#).

Definition 3.13. A *mechanism-purview pair* is a pair of subsystems of a given system $(S, \mathbb{D}, T)$, depending on the same global state $s: \mathbb{1} \rightarrow S$.

Intuitively a mechanism describes the subset of S that we are “letting make a difference”, and the purview describes the subset of S that is affected.

Definition 3.14. Given a mechanism $(M, \mathbb{D}_M, T_M)$ and a purview $(P, \mathbb{D}_P, T_P)$, we define the *cause* to be

process

$$\text{cau}: M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P,$$

where M' and P' are the complements in the decompositions defined by M and P respectively. Similarly, the *effect* is defined as

$$\text{eff}: M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T^\dagger} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P.$$

Note that both *cau* and *eff* send causal states to causal states. Intuitively, these capture the way the previous and next state of P is constrained by the restricted state $s|_M$. When the mechanism is the unit $\mathbb{1}$, then the cause and effect *cau, eff*: $\mathbb{1} \rightarrow P$ are respectively called the *unconstrained cause*, denoted *ucau* and the *unconstrained effect*, denoted *ueff*.

Definition 3.15. Given a state s and a mechanism-purview pair M, P , the associated *cause repertoire* is defined to be the state

$$\text{cau}(s, M, P) = \mathbb{1} \xrightarrow{s|_M} M \xrightarrow{\text{cau} \otimes \text{ucau}} P \otimes P' \simeq S.$$

One defines an *effect repertoire* *eff*(s, M, P) similarly.

We also need to decompose and cut causes and effects.

Definition 3.16. Given a decomposition (M_1, M_2) of a mechanism M , and a decomposition (P_1, P_2) of a purview P , the *decomposed cause* is the process

$$\text{cau}_{M_1, M_2}^{P_1, P_2}: M \simeq M_1 \otimes M_2 \xrightarrow{\text{cau} \otimes \text{cau}} P_1 \otimes P_2 \simeq P,$$

and again similarly for defining the *decomposed effect*.

The *decomposed cause* and *effect repertoires* are then defined similarly to the normal ones, by replacing *cau* and *eff* with their decomposed variants in the definitions. We denote these states by $\text{cau}_{M_1, M_2}^{P_1, P_2}(s, M, P)$ and $\text{eff}_{M_1, M_2}^{P_1, P_2}(s, M, P)$ respectively. Given a cut direction α , the *cut cause* and *effect repertoires* $\text{cau}^\alpha(s, M, P)$ and $\text{eff}^\alpha(s, M, P)$ is similarly defined by using the cut time evolution $T^{\psi, \alpha}$, where ψ is the decomposition associated to the mechanism M . Combining the two approaches yields also *cut-decomposed cause* and *effect repertoires*.

For a generalized IIT, one needs to be able to compare causes and effects along different substructures. This is done by assuming that there is a distance function

on the states of S : $d(-, -): \text{St}(S) \times \text{St}(S) \longrightarrow \mathbb{R}^+$. As in [CITE](#), we do not require this to be a metric, though it usually will be in examples. In classical IIT, this metric is the Wasserstein metric, while in quantum-classical IIT it is the metric induced by the norm of the system C^* -algebra.

We now have enough data to define the IIT-algorithm, as presented in [CITE](#) and [CITE](#). The algorithm defines integration for all cause and effect repertoirs, and uses this to aggregate upwards to an integration value for the total system S .

Definition 3.17. Let $\text{cau}(s, M, P)$ be a cause repertoire. Its *integration* is defined as the minimum distance from $\text{cau}(s, M, P)$ to a decomposed repertoire associated to a non-trivial decomposition:

$$\varphi(\text{cau}(s, M, P)) := \min_{(M_1, M_2), (P_1, P_2)} d(\text{cau}(s, M, P), \text{cau}_{M_1, M_2}^{P_1, P_2, \text{cut-dynamics } T|_U^{\psi, \alpha}}(s, M, P)).$$

We define the integration level of an effect repertoire $\text{eff}(s, M, P)$ similarly.

Definition 3.18. Given a state $s: \mathbb{1} \longrightarrow S$ and a sub-system $M \in \text{Sub}_s(S)$, the *core cause* is the sub-system $P^{cc} \in \text{Sub}_s(S)$ such that the associated cause repertoire $\text{cau}(s, M, P)$ has maximal integration. Similarly, the *core effect* is the sub-system P^{ce} that maximizes the integration of the associated effect repertoire $\text{eff}(s, M, P)$.

Definition 3.19. The integration of a mechanism M is then defined as

$$\varphi(M) := \min(\varphi(\text{cau}(s, M, P^{cc})), \varphi(\text{eff}(s, M, P^{ce}))).$$

Given a sub-object $M \in \text{Sub}_s(S)$, the triple consisting of the core cause repertoire, the core effect repertoire and the resulting integration defines the *concept* of M . Allowing M to vary, defines the *Q-shape* of the state s . The collection of all *Q-shapes* defines the *experience space* $\mathbb{E}(S)$ of the system S . See [CITE](#) for the details.

Remark 3.20. One of the fundamental beliefs in IIT is that any conscious experience a system S can have is an element of $\mathbb{E}(S)$.

By minimizing over all possible decompositions, one can similarly define the integration of a *Q-shape*. The subsystem with maximally integrated *Q-shape* is the system's *major complex*, which by definition defines an element $\mathbb{E}(S, s)$ in the experience space $\mathbb{E}(S)$. IIT

claims that this element is the particular conscious experience of the system S in the state s , and the associated Φ -value is defined as the size of this conscious experience: $\Phi(S, s) = \|\mathbb{E}(S, s)\|$.

If this Φ -value is zero, then it means that for all mechanisms M , the integration $\varphi(M)$ is zero, meaning that for all purviews P there is a decomposition with the same integration of its cause and effect repertoire. In principle, decomposing the system does not change its cause and effect repertoirs. We now use this to prove our main result.

Lemma 3.21. Let $(S, \mathbb{D}, T)$ be a system, $s: \mathbb{1} \longrightarrow S$ a state of S , (A, B, ψ) a decomposition and α a cut direction. If $\Phi(S, s) = 0$, then for every sub-system $U \in \text{Sub}_s(S)$ the restricted dynamics $T|_U$ and the restricted

cut-dynamics $T|_U^{\psi, \alpha}$ are equal in $\mathcal{F}(U)$.

Proof. By definition there is a complement U' of U such that (U, U') is a decomposition of S . We consider U as a mechanism, with purview the total system S . The associated cause repertoire $\text{cau}(s, U, S)$ is equivalent to the composite

$$\mathbb{1} \xrightarrow{s|_U} U \xrightarrow{T|_U} U \xrightarrow{1_U \otimes w_{U'}} U \otimes U' \cong S.$$

The same description holds for the cut-decomposed cause repertoire $\text{cau}_{U, U'}^{\mathbb{1}, S, \alpha}(s, U, S)$, i.e., it is equivalent to the composite

$$\mathbb{1} \xrightarrow{s|_U} U \xrightarrow{T|_U^{\psi, \alpha}} U \xrightarrow{1_U \otimes w_{U'}} U \otimes U' \cong S.$$

As the embedding $\text{End}(U) \hookrightarrow \text{End}(U)$ defined by $f \mapsto (\text{Id}_U \otimes w_{U'}) \circ f \circ s|_U$ is injective in the image $\mathcal{F}(U)$, equality of the cause repertoirs implies equality of the underlying endomorphisms of U . In particular, the restricted endomorphisms $T|_U$ and $T|_U^{\psi, \alpha}$ are equal if and only if the associated cause repertoirs are equal.

Now, by assumption we have $\Phi(S, s) = 0$, which by definition means we have non-trivial decompositions of U and S such that either the cause or effect repertoire factorizes. Without loss of generality we assume this to be the cause repertoire: $\text{cau}(s, U, S) = \text{cau}_{U_1, U_2}^{S_1, S_2}(s, U, S)$. By construction, the cut time evolutions and the cause and effect maps are defined by inserting noise on complements. Hence, this equivalence on cause repertoirs implies that the action of T on U is the same as the restricted action of $T^{\psi, \alpha}$.

Given a 1-simplex $\sigma: U \rightarrow V$ we then have $\mathcal{F}(\sigma)(T) = \mathcal{F}(T^{\psi,\alpha})$, which by definition implies that $T|_U$ and $T|_U^{\psi,\alpha}$ are equivalent in $\mathcal{F}(U)$. $\square$

Remark 3.22. Injectivity in the above argument relies crucially on the use of the noise state w rather than the actual system state. In [Remark 3.5](#) we mentioned the option of state-dependent integration, but in this case the above argument fails, and one would need a different approach to make similar claims.

Proof of Theorem 3.12. The assumption of vanishing integrated information $\Phi(S, s) = 0$ implies by definition that for all mechanisms M , we have

$$\varphi(M) = \min(\varphi(\text{cau}(s, M, P^{cc})), \varphi(\text{eff}(s, M, P^{ce}))) = 0.$$

In particular, for all mechanism-purview pairs M, P , there exists decompositions $M \simeq M_1 \otimes M_2$ and $P \simeq P_1 \otimes P_2$ such that either

$$\text{cau}(s, M, P) = \text{cau}_{M_1, M_2}^{P_1, P_2}(s, M, P),$$

or

$$\text{eff}(s, M, P) = \text{eff}_{M_1, M_2}^{P_1, P_2}(s, M, P).$$

The goal is to exploit this fact to show that all obstruction classes are coboundaries. To accomplish this, we fix a decomposition $\psi: S \simeq A \otimes B$ and a cut direction $\alpha \in \{\rightarrow, \leftarrow, \leftrightarrow\}$. To show that the associated obstruction class $[\Delta^{\psi,\alpha}] = 0$ it suffices to show that it is a coboundary of some 0-cochain θ , in other words that $\Delta^{\psi,\alpha} = \delta\theta$.

We define this cochain by $\theta(U) := T|_U$ for all $U \in \text{Sub}_s(S)$. The coboundary is given on a 1-simplex $\sigma: U \rightarrow V$ by

$$\delta\theta(\sigma) = \mathcal{F}(\sigma)(T|_V) - T|_U.$$

By [Lemma 3.21](#) we have $\mathcal{F}(\sigma)(T|_V) = \mathcal{F}(\sigma)(T|_V^{\psi,\alpha}) = T|_U^{\psi,\alpha}$, hence $\delta\theta(\sigma) = -\Delta^{\psi,\alpha}(U)$. In particular, $\Delta^{\psi,\alpha} = -\delta\theta$, so the obstruction is a coboundary, meaning that $[\Delta^{\psi,\alpha}] = 0$ in the integrated cohomology group $H_{int}^1(S)$. $\square$

4 Extensions

4.1 Cohomological integrated information

4.2 Higher order integration